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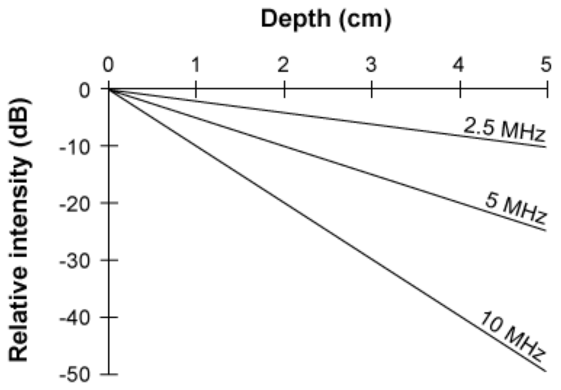


Figure 1 Penetration in soft tissue for three different ultrasound frequencies

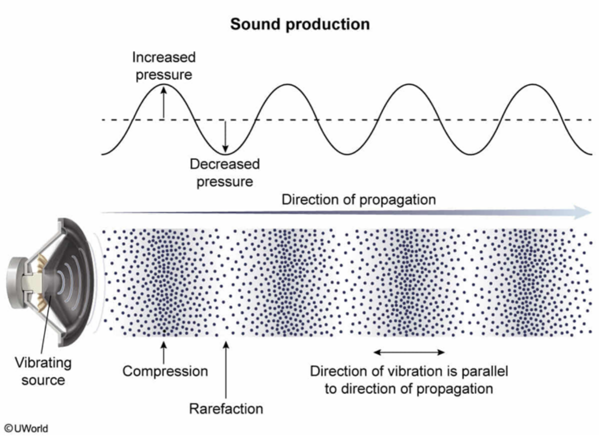
Sound waves propagate through soft tissue at an average speed of 1,500 m/s and through blood at 1,570 m/s.

Which of the following is NOT a property of sound?

- A. It attenuates most in soft materials. [15%]
- B. It is fastest in a vacuum. [62%]
- C. It is a longitudinal wave. [6%]
- D. It is a pressure wave. [15%]

Correct
63% answered correctly

Explanation:



Sound propagates through the **vibrations of molecules** in a medium. This vibration creates oscillations of compressions (areas of high pressure) and rarefactions (areas of low pressure), resulting in a propagating **pressure wave (Choice D)**. Sound is a longitudinal wave; the molecules vibrate back and forth parallel to the direction of propagation **(Choice C)**.

Attenuation (damping) is the decrease in the amplitude (intensity) of a wave due to absorption and scattering. The magnitude of the attenuation depends on the properties of the medium; attenuation is **greater for softer materials (Choice A)**. Sound is also attenuated as a function of distance traveled by the inverse-square law. Intensity is inversely proportional to the square of the distance from the source.

The **speed** of sound is slowest in gases, faster in liquids, and **fastest in solids**. Because sound propagates through the vibrations of the molecules in a medium, sound **cannot propagate through a vacuum**.

Educational objective:

Sound propagates through the vibrations of the molecules as longitudinal pressure waves, and therefore cannot exist in a vacuum. The attenuation of sound is greatest in soft materials and increases with distance. Sound travels most slowly in gases and most quickly in solids.

Time spent:0 sec
Subject:
Physics

QID:400920
System:
Light and Sound

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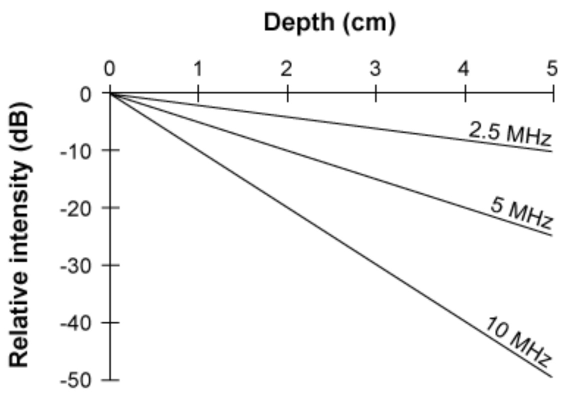

Figure 1 Penetration in soft tissue for three different ultrasound frequencies

Sound waves propagate through soft tissue at an average speed of 1,500 m/s and through blood at 1,570 m/s.

How much farther will a 5 MHz signal penetrate compared to a 10 MHz signal when the initial intensity of each signal decreases by a factor of 100?

- A. 1 cm [16%]
- B. 2 cm [53%]
- C. 4 cm [15%]
- D. 10 cm [13%]

Correct
53% answered correctly

Explanation:

Decibel scale

Decibel (dB)	Relative intensity	Perceived loudness
−20	×1/100	×1/4
−10	×1/10	×1/2
0	×1	×1
+10	×10	×2
+20	×100	×4

Intensity is the amount of power (energy per unit time) delivered per unit area. In Figure 1, the relative intensity of ultrasound signals is graphed using the decibel scale. The **decibel scale** relates the perceived loudness of a sound to its actual intensity. The perception of sound is approximately **logarithmic**: a sound that is 10 times more intense is twice as loud.

The ratio of the attenuated signal intensity I to the source intensity I_S in decibels (dB) is given by:

$$[\mathrm{dB}] = 10\log\left(\frac{I}{I_S}\right)$$

A simple relationship can be used to relate dB changes to relative intensities: a 10-fold increase in intensity is a change of +10 dB; similarly, a **10-fold decrease** in intensity is a change of **−10 dB**. Therefore, a **100-fold decrease in intensity** corresponds to a change of (−10) + (−10) = **−20 dB**. Alternatively, this relationship can be resolved mathematically.

From Figure 1, the 10 MHz and 5 MHz signals are attenuated by 20 dB at a depth of **2 cm** and **4 cm**, respectively. The difference between the two depths is:

$$4\ \mathrm{cm} - 2\ \mathrm{cm} = 2\ \mathrm{cm}$$

Educational objective:
The decibel (dB) scale is logarithmic and relates the perceived loudness of a sound to its actual intensity. For each 10-fold decrease, sound intensity decreases by 10 dB. Therefore, a 100-fold decrease in sound intensity corresponds to a decrease by 20 dB.

Time spent:0 sec
Subject:
Physics

QID:400921
System:
Light and Sound

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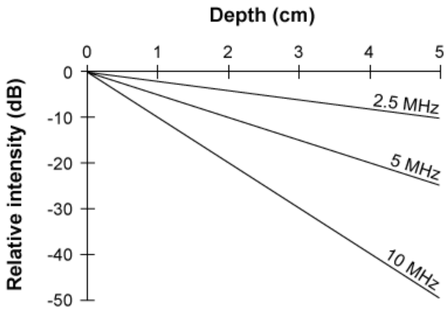
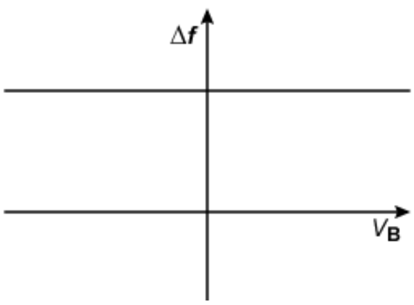


Figure 1 Penetration in soft tissue for three different ultrasound frequencies

Sound waves propagate through soft tissue at an average speed of 1,500 m/s and through blood at 1,570 m/s.

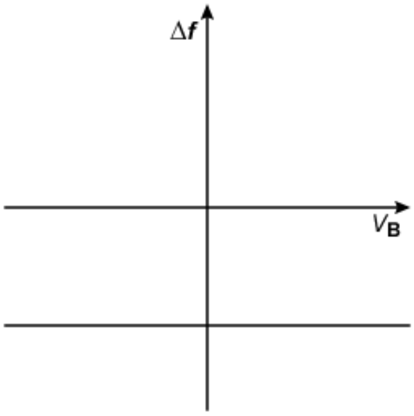
Blood flows toward and later passes by a doppler ultrasound probe that is fixed in place. Relative to the probe, the component of the blood velocity v_B along the line connecting the blood and the probe changes from negative to positive as the blood passes by the probe. Which of the following best illustrates the relationship between v_B and the frequency shift of returning waveforms Δf?

A.



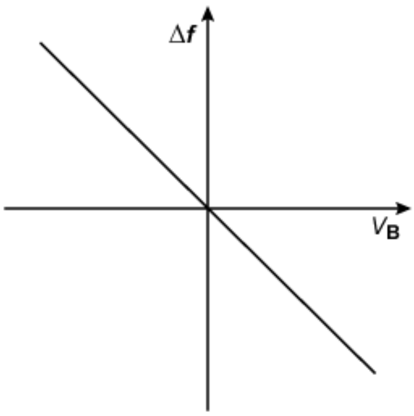
[12%]

○ B.



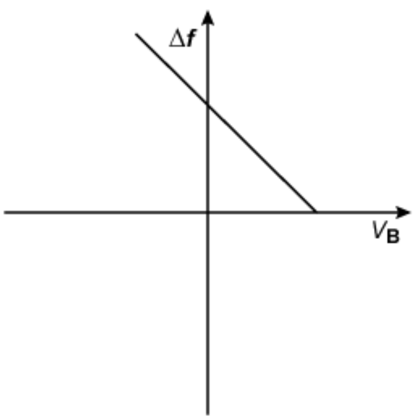
[8%]

✔ ◉ C.



[56%]

○ D.

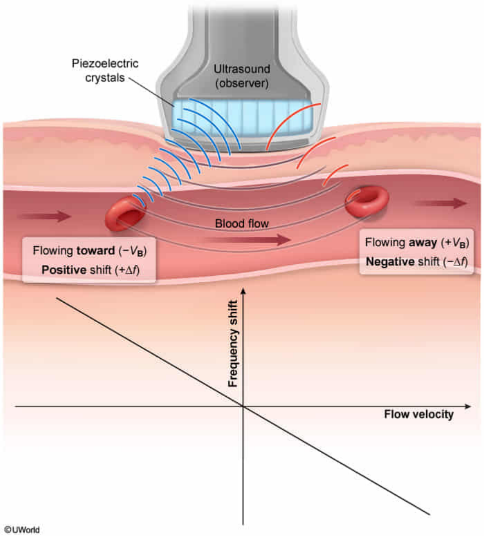


[22%]

Correct
56% answered correctly

Explanation:

Explanation:



With relative motion between a source and an observer, the **observed waveform differs** from the original, emitted waveform in terms of both frequency and wavelength. This phenomenon is known as the Doppler effect, and the frequency shift is proportional to the component of the source velocity along a line between the source and the observer.

According to the passage, blood flow velocity is determined by the magnitude of the Doppler shift (Δf) detected by the **ultrasound device** (ie, the observer). Because the ultrasound waves are first reflected off the blood before detection by the probe occurs, the flowing **blood** acts as the source of sound.

When blood is flowing toward the device, the blood velocity has a component that is directed towards the ultrasound probe. Hence, v_B is **negative** (ie, *subtracted* from waveform velocity) and the **frequency shift is positive** (ie, observed frequency > original frequency). A positive frequency shift occurs because each successive wave is reflected closer to the device.

Conversely, when the blood has passed by the device, a component of its velocity is directed away from the probe. Hence, v_B is **positive** (ie, *added* to waveform velocity) and the **frequency shift is negative** (ie, observed frequency < original frequency). A negative frequency shift occurs because each successive wave is reflected farther from the device.

(Choice A) A positive frequency shift means that observed waveform frequency increases. An increase in received waveforms would only occur as blood flows toward the ultrasound device (ie, negative velocity).

(Choice B) A negative frequency shift means that observed waveform frequency decreases, which would only occur as blood flows away from the ultrasound device (ie, positive velocity).

(Choice D) The plot must cross the x-axis because the frequency shift will change from positive to negative as blood transitions from approaching the probe to moving away from the probe.

Educational objective:
The Doppler effect occurs when the observed frequency and wavelength of a sound are shifted from those of the original due to relative motion between the source and the observer. The frequency shift is positive when the source velocity is negative (moving closer) and negative when the source velocity is positive (moving away).

Time spent:0 sec
Subject:
Physics

QID:400922
System:
Light and Sound

Ultrasound is a technique that uses the propagation properties of high-frequency sound waves to construct images of internal organs or to measure blood flow velocity. The shape and position of internal organs are resolved by measuring the return time of reflected ultrasound waves. Similarly, blood flow velocity is resolved by measuring the Doppler shift of reflected ultrasound waves. An ultrasound device consists of an array of piezoelectric crystals and an acoustic lens. The piezoelectric crystals generate sound when an alternating voltage is applied, and the acoustic lens increases the transmission efficiency from the device to the body by reducing sound reflection at the skin.

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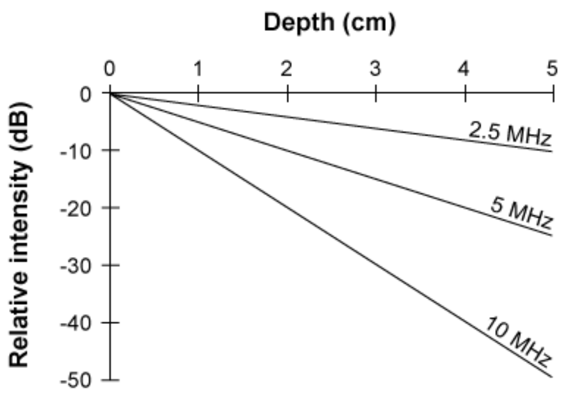


Figure 1 Penetration in soft tissue for three different ultrasound frequencies

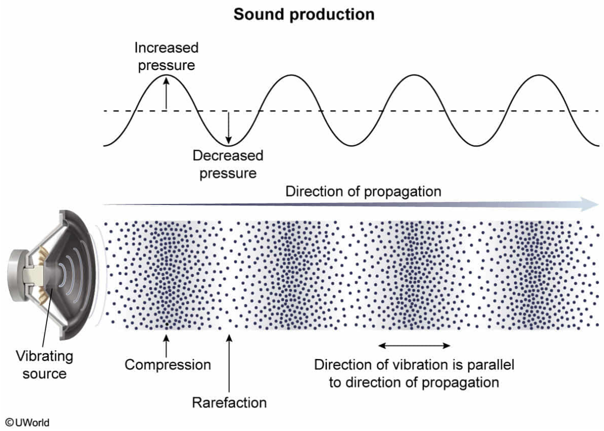
Sound waves propagate through soft tissue at an average speed of 1,500 m/s and through blood at 1,570 m/s.

Piezoelectric crystals likely create sound by:

- A. rapidly expanding and contracting. [43%]
- B. emitting high-energy photons. [10%]
- C. generating heat waves. [1%]
- D. sending pulses of electricity. [43%]

Correct
43% answered correctly

Explanation:



Sound is propagated in the form of longitudinal waves of oscillating pressure. **Compressions** (areas of high pressure) and **rarefactions** (areas of low pressure) are formed through the back-and-forth **vibrations** of the molecules of the medium.

The passage states that piezoelectric crystals produce sound when an alternating voltage is applied to them. It is reasonable to conclude that the **expansion and contraction** of the crystals will cause molecules in the surrounding medium to **vibrate**, creating **sound waves**.

(Choice B) A photon is a packet of energy that makes up electromagnetic radiation (light). The emission of photons would generate light, not sound.

(Choices C and D) Sound energy is propagated in the form of pressure waves, not heat waves or electric waves.

Educational objective:

Sound is propagated in the form of pressure waves by the vibrations of the molecules in a medium. The rapid expansion and contraction of crystals creates pressure waves (sound) by vibrating nearby particles.

Time spent:0 sec
Subject:
Physics

QID:400923
System:
Light and Sound

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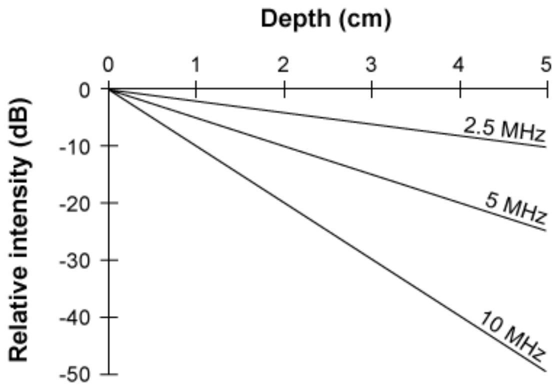


Figure 1 Penetration in soft tissue for three different ultrasound frequencies

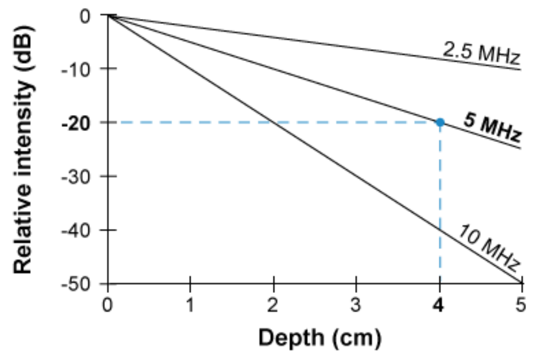
Sound waves propagate through soft tissue at an average speed of 1,500 m/s and through blood at 1,570 m/s.

What is the period of an ultrasound signal that is attenuated by 20 dB after it travels 4 cm into soft tissue?

- A. 5×10^{-9} s [5%]
- ✔ B. 2×10^{-7} s [48%]
- C. 5×10^{-6} s [32%]
- D. 2×10^{-4} s [12%]

Correct
49% answered correctly

Explanation:



The **period** *T* of a wave is the **amount of time for one cycle** (or one wavelength) to pass through a fixed point. Period is the **reciprocal (inverse) of frequency**:

$$T = \frac{1}{f}$$

Frequency has units of hertz (Hz), which are equivalent to inverse seconds (s^{-1}); therefore, period has units of seconds. From Figure 1, a **5 MHz** sound wave penetrates 4 cm before it is attenuated (decreased) by 20 dB. A frequency of 5 MHz is equal to 5×10^6 Hz. Using this value to calculate the period gives:

$$\frac{1}{5 \times 10^6 \text{ s}^{-1}} = 0.2 \times 10^{-6} \text{ s} = 2 \times 10^{-7} \text{s}$$

Educational objective:

The period of a wave is the amount of time for one cycle to pass through a fixed point, and it is the reciprocal of frequency: *T* = 1/*f*.

Time spent:0 sec
Subject:
Physics

QID:400924
System:
Light and Sound

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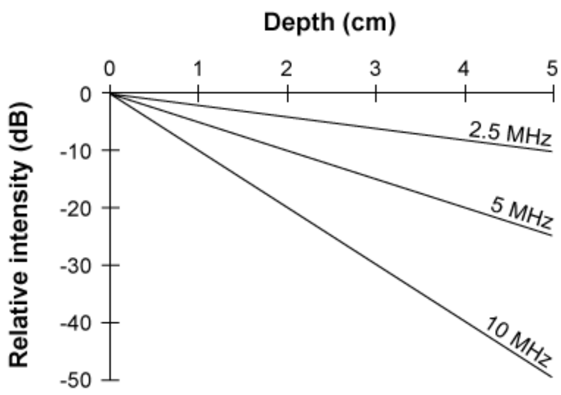


Figure 1 Penetration in soft tissue for three different ultrasound frequencies

Sound waves propagate through soft tissue at an average speed of 1,500 m/s and through blood at 1,570 m/s.

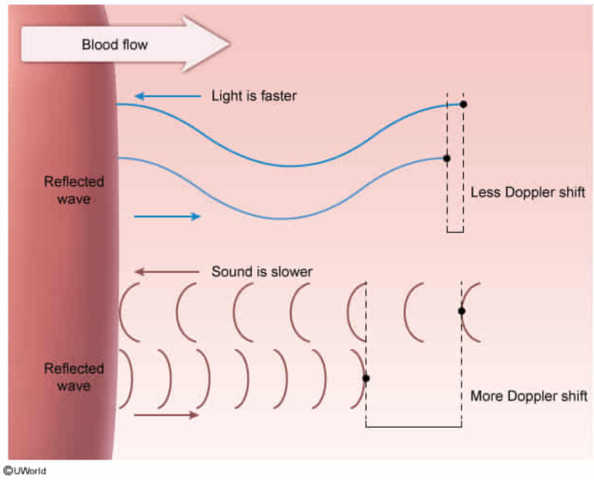
Laser Doppler flowmetry uses an infrared laser to measure blood flow velocity. Compared to ultrasound, the percent change in frequency using this technique is:

- A. smaller. [47%]
- B. the same. [18%]
- C. larger. [29%]
- D. zero. [3%]

Correct
48% answered correctly

Explanation:

Effect of wave speed on Doppler shift



Given relative motion between a source and an observer, the observed wave differs from the original wave in frequency and wavelength. This phenomenon is known as the Doppler effect and the percent change in the frequency can be approximated by:

$$\frac{\Delta f}{f} = \frac{v}{c}$$

where Δf is the frequency shift, f is the original frequency, v is the relative velocity between the source and the observer, and c is the speed of the wave in the medium. From the above relationship, the **percent change in frequency** is **inversely related** to the **speed of the wave**.

An infrared laser uses a form of electromagnetic radiation (light), which travels at the speed of light $c = 3 \times 10^8$ m/s. The speed of sound in blood is given in the passage as c = 1,570 m/s. Because the **speed of light** is much **greater than** the **speed of sound**, the observed percent change in the frequency **will be smaller** for laser Doppler flowmetry.

Educational objective:
The Doppler effect can be approximated by the relative velocity between the source and the observer and the speed of the wave in the medium $\frac{\Delta f}{f} = \frac{v}{c}$. The frequency shift in percent is inversely proportional to the wave's speed. The Doppler effect of light will result in a smaller percent change in frequency because the speed of light is much greater than that of sound.

Time spent:0 sec
Subject:
Physics

QID:400925
System:
Light and Sound